

BUSINESS MASTER OF SCIENCE CAPSTONE PROJECT

PROJECT INITIATION DOCUMENT

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PROJECT SPONSOR	
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CAPSTONE PROJECT MANAGER	
Name	Eduardo Arino de la Rubia
PROJECT DETAILS	
Title	
Power Sector Transition Tracker for Uzbekistan: Data-Driven Insights for ILF's Energy Advisory Work	
Background (summary of reasons for project sponsor's interest in/significance of project)	
ILF Consulting Engineers Austria GmbH has long-standing advisory engagement in Uzbekistan's energy sector, working with the Ministry of Economy and Finance, the National Energy Efficiency Agency, and IFIs including the World Bank, EBRD, and ADB. Uzbekistan's power sector is under simultaneous structural pressure: rapidly rising electricity demand, declining domestic gas supply forcing an accelerated energy transition, expanding renewables constrained by grid capacity, severely ageing transmission and distribution infrastructure, and unprecedented levels of private and multilateral investment flows. These pressures create a multi-year pipeline of ILF advisory opportunities across generation, grid modernization, renewable integration, energy efficiency, and PPP advisory. Identifying and monitoring these opportunities currently requires manual effort across fragmented data sources, which this project addresses by building a consolidated, annually refreshable Power Sector Transition Tracker.	
Statement of key problems, objectives and desired outcomes	
ILF has no consolidated tool linking Uzbekistan's electricity supply, demand, emissions, grid performance, and investment signals in one annually refreshable view. The supply-demand gap is widening, grid bottlenecks are unquantified in business terms, declining gas production is creating unplanned transition pressure, and renewable resource potential is not spatially linked to grid constraints. The objective is to build a reproducible tracker — a Python data pipeline, forecasting models, and interactive Plotly Dash dashboard — consolidating multi-source data from 1990 to 2026, forecasting the system to 2040 under several scenarios, and generating structured investment signals mapped to ILF's opportunity categories. Desired outcomes include a clean master dataset, exploratory analysis, forecasting models for demand, renewable penetration, fossil decline and emissions, an investment signal panel, a dashboard with regional spatial map, and a written ILF advisory summary.	

PROJECT DETAILS
Interim and final deliverables (incl. format of project presentation and technical discussion, target audience)
The project delivers a Python data pipeline, four Jupyter notebooks covering exploratory analysis, forecasting, investment signals, and spatial analysis, and a Plotly Dash dashboard with scenario controls, a Plan B nuclear scenario overlay toggle (enabling users to activate a parametric nuclear capacity addition for sensitivity analysis), and regional map. Final outputs are a deployed dashboard, a GitHub repository with full documentation, and a short ILF written summary in opportunity table format. Interim notebooks are shared with the sponsor at each phase review, and the dashboard is demonstrated live in Phase 7. Target audience is ILF's advisory and business development teams, including non-technical stakeholders.
Methods of approach (e.g. data, information resources, software, methodologies, key contacts)
Data is drawn from six open-licensed sources covering 1990–2026: IEA, IRENA, World Bank, Uzbekistan Statistics Committee, EDB March 2026 report, and Global Solar and Wind Atlases — approximately 65 files in total. Software includes Python, Plotly Dash, Geopandas, and Jupyter Notebooks, with GitHub for version control and Render or HuggingFace Spaces for deployment. Methodologies cover exploratory statistical analysis, ARIMA and OLS demand forecasting, renewable penetration and fossil decline modelling, CO₂ forecasting, a parametric Plan B nuclear overlay, threshold-based investment signal detection mapped to ILF's five opportunity categories, and spatial analysis combining oblast-level data with GeoJSON and resource atlas layers. Predictive Modelling Component <i>Target variables:</i> The primary target variables are total electricity demand (TWh), renewable energy share (% of generation), fossil-based electricity generation (TWh), and CO₂ intensity of the power sector (gCO₂/kWh). These are forecast annually from 2026 to 2040. <i>Modelling approach:</i> Two complementary methods are applied. OLS regression models demand as a function of GDP, population, and industrial output, capturing long-run structural drivers. ARIMA time-series models are used to forecast renewable penetration and fossil generation trajectories, with scenario extrapolation applied under several energy-mix assumptions extending to 2040. <i>Evaluation framework:</i> Model performance is assessed on held-out test data using Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and R² (coefficient of determination). These metrics are reported for each target variable and each model variant to allow transparent comparison across approaches. <i>Data structure and validation plan:</i> The historical dataset covers 1990–2026. A chronological 80/20 train-test split is applied, using 1990–2018 for training and 2019–2026 for testing, preserving temporal order. Time-series cross-validation with an expanding window is used during model selection to avoid data leakage and ensure robustness across different time periods.
Assumptions (e.g. expected input from client, possible constraints, expected location of work, exclusions)
If ILF provides a Data360 dataset, it becomes the primary base. All sources are open-licensed, nominal GDP is deflated before any modelling, and preliminary 2024–2026 figures are clearly labelled throughout all outputs. Nuclear is treated as Plan B rather than baseline given recently commenced construction, no Central Asian commissioning precedent, and Rosatom uncertainty. The dashboard is a working prototype; production-grade deployment, real-time feeds, financial modelling, and primary data collection are out of scope.

INITIAL HIGH-LEVEL PROJECT PLAN AND SCHEDULE

Work breakdown and timing
Phase 1 (6–16 March) — project understanding. Phase 2 (16–22 March) — PID drafted and reviewed at kick-off on 22 March. Phase 3 (22 March – 10 April) — data pipeline construction, analysis plan, and initial exploratory analysis (beginning substantive analytical work immediately alongside pipeline setup). Phase 4 (10–24 April) — full exploratory analysis. Phase 5 (24 April – 4 May) — forecasting models. Phase 6 (4–11 May) — investment signals and interim report. Phase 7 (11–25 May) — dashboard development and demo. Phase 8 (25 May – 8 June) — final report and submission.
Quality review procedure (review meetings)
Review 1 — 22 March. Review 2 — 3 April. Review 3 — 17 April. Review 4 — 1 May. Review 5 — 15 May. Review 6 — 29 May. Review 7 — 5 June.
Anticipated expenses if any (must be covered by project sponsor or client organization)
No expenses are anticipated. All data, software, and hosting are free and open-licensed. No expenses are expected to be covered by the project sponsor or client organisation.

STUDENT

Name	Date	Signature (add name if sending by email)
Farangiz Jurakhonova		

PROJECT SPONSOR

Name	Date	Signature (add name if sending by email)
Ardak Akhatova		

CAPSTONE PROJECT MANAGER

Name	Date	Signature (add name if sending by email)
Eduardo Arino de la Rubia		

BUSINESS MASTER OF SCIENCE CAPSTONE PROJECT

LETTER OF TERMS

In order to ensure that the Business Master of Science Capstone Project (the Project) of Central European University (the University) runs smoothly and to mutual advantage, please note the following terms under which the project is undertaken.

- The Project is undertaken by the Student as part of his or her studies at the University. The student is not an employee of the University, so the University cannot enter agreements on his or her behalf or take any liability for his or her actions.
- The Project is undertaken as part of an educational program and is supervised and examined by a Faculty Supervisor appointed by the University. The Faculty Supervisor is bound contractually and by law to keep confidential any confidential information disclosed to him or her in the supervision and examination of the Project.
- The client organization (the Client) must assign a named Project Sponsor, who will be available over the duration of the project to advise and guide the Student. The Project Sponsor is expected to give feedback to the Faculty Supervisor on the conduct of the work.
- The Student and the Faculty Supervisor sign the General Confidentiality and Non-disclosure Agreement (NDA). The Project Sponsor may request a separate confidentiality and non-disclosure agreement to be signed.
- The Project is a student educational project and should be seen in this context. The work does not constitute professional advice, and no warranties are made regarding the information presented. Neither the Student nor the University and its faculty accept any liability for the consequences of any action taken as a result of the work, or any recommendations made or inferred.
- The intellectual property rights to the work undertaken and/or the deliverables produced vest in the Client.
- The Student is required to declare his or her association with the University and with the Client when collecting information from other organizations.
- The Project Sponsor must provide support, guidance, and opportunities for consultation to enable the student to carry out the Project effectively.
- For the purpose of the present agreement the Project Sponsor is not a representative of the Client. The Project Sponsor is agreeing to the terms above as an individual and the agreement does not affect the Project Sponsor's affiliation with the Client in any way.
- Expenses incurred in the execution of the Project must be met by the Client and paid directly to the Student.

Please confirm that you have read and agreed the terms of engagement by signing below.

PROJECT SPONSOR

Name / client organization	Date	Signature (add name if sending by

BUSINESS MASTER OF SCIENCE CAPSTONE PROJECT

GENERAL CONFIDENTIALITY AND NON-DISCLOSURE AGREEMENT

During the course of the Business Master of Science Capstone Project (the Project) of Central European University (the University), I am likely to discuss and have access to information, technology and ideas that various organizations participating in the program regard as confidential.

I agree for the benefit of the client organization (the Client), the University and other participating organizations that I will make all reasonable efforts to hold in strict confidence any information, technologies and ideas that I am told in advance are confidential, that I will not copy, reveal or disclose such information, technology and ideas to any third party, and that I will not use any such information, technology and ideas for my own benefit or for the benefit of any organization with which I am affiliated now or in the future. I also agree to abide by such other rules and guidelines that the University may reasonably impose.

This Agreement shall not apply to any information, technology and ideas which:

- (1) at the date of this Agreement are in the public domain or subsequently come into the public domain through no fault of mine;
- (2) were already known to me on the date of disclosure, provided that such prior knowledge can be substantiated;
- (3) properly and lawfully become available to me from sources independent of the supplying party;
- (4) are disclosed pursuant to the requirement or request of a governmental agency provided that in such an event I shall inform you of the nature and extent of any disclosure so required.

This Agreement shall come into effect from the date below, and the obligation under the Agreement shall remain in effect indefinitely unless agreed otherwise in writing by the Client. This Agreement shall be subject to Austrian law.

STUDENT

Name	Date	Signature (add name if sending by
Farangiz Jurakhonova		

CAPSTONE PROJECT MANAGER

Name	Date	Signature (add name if sending by
Eduardo Arino de la Rubia		

PROGRAM COORDINATOR

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CEU PRO-RECTOR FOR RESEARCH

Name	Date	Signature (add name if sending by
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PROGRAM HEAD (IF APPLICABLE)

Name	Date	Signature (add name if sending by
Gabor Bekes		

FACULTY SUPERVISOR (IF APPLICABLE)

Name	Date	Signature (add name if sending by

